

File Watcher & Automated Comparison Report System

Developer Requirements Document

MVP - Phase 1

Client: Quenton

Platform: ASP.NET on IIS / SQL Server

Industry: Mining / Personnel Compliance

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1. Project Overview

Build an automated File Watcher and Comparison Report System for a mining company with 3 sites. The system monitors a designated network folder for incoming Excel files (Evac Reports and AlcoConnect Breathalyser Reports), automatically imports their data into a SQL Server database, and generates comparison reports to identify personnel compliance gaps.

The core purpose is to reconcile two data sources - an evacuation roster (who should be on site) and breathalyser test results (who was actually tested) - to instantly see:

- Who is on the roster but has NOT been tested (compliance gap)
- Who was tested but is NOT on the roster (unmatched records)
- Who appears in both files (fully compliant / matched)

User Workflow:

1. Select Report Date -> 2. Select Site -> 3. View Results in 3 Groups -> 4. Export if needed

Key Constraints:

- 3 mining sites, each producing their own set of files daily
- Evac Report: 1 file per site per day (uploaded once, ~150 rows)
- AlcoConnect Report: Multiple files per site per day (up to 12 arrivals)
- No manual upload interface needed - automated folder monitoring only
- Duplicate records in multi-import files must be deduplicated automatically (keep first version)

2. Data Sources & File Specifications

2.1 Evac Report (Master File)

One file per site per day. This is the roster of personnel who are on-site and should be tested. File naming pattern: "DAL Daily Evac Report - AlcoConnect(DD-MM-YYYY).xlsx"

Columns:

Column Name	Description	Example
workgroup	Employee workgroup assignment	DAL Mining UG - Barmingo
Name	Employee full name (Last, First)	BROWN, James
Organisation	Company / contractor name	Barmingo
RosterDate	Date of the roster entry	2026-03-16
Work Site	Mining site name	Dalgaranga
WorkStatus	Current shift/status	Fly In PM - Nightshift
Room	Accommodation room assignment	DAL B-23
mobile	Employee mobile number	0410 903 640
IMAOOpenINX	Linking ID (contains letters + number)	RAM-a25099A

2.2 AlcoConnect / Breathalyser Report

Multiple files per site per day (up to 12 arrivals throughout the day). These contain breathalyser test results. File naming pattern: "breathalyser_activity_report_DDMonYYYY_XXXXXXXXXX.xlsx"

Columns:

Column Name	Description	Example
Site	Mining site name	Dalgaranga Operations
Staff ID	Numeric staff identifier	01071980
Staff Name	Employee name	Mark CHERUBINO
Job Title	Employee job title	(may be blank)
Phone	Phone number	(may be blank)
Email	Email address	(may be blank)

Manager	Manager name	(may be blank)
Machine Type	Breathalyser device model	WM4
Date	Test date (DD-MM-YYYY)	16-03-2026
Time	Test time (HH:MM:SS)	05:11:17
Result	BAC test result	0.000 g/100mL BAC
Serial Number	Device serial number	33000868
Location	Test location on site	Barmingo

2.3 Record Linking Logic

The two files are linked by employee ID, but the format differs between files:

File	Field Name	Format	Example
Evac Report	IMAOpenINX	Letters + Number + Letter (e.g., RAM-x12345Y)	RAM-a25099A
AlcoConnect	Staff ID	Pure numeric (may have leading zeros)	01071980

Matching Rule:

Strip all non-numeric characters from the Evac IMAOpenINX field to extract the numeric ID, then match against the AlcoConnect Staff ID (also stripped of leading zeros if needed). Example: "RAM-a25099A" -> extract "25099" and match against Staff ID "25099".

3. System Architecture

Component	Technology	Details
Web Application	ASP.NET (Core or Framework)	Hosted on existing IIS server; Date/Site selection + 3-group report display
Database	SQL Server	Normalized schema with stored procedures for data processing and comparison
File Watcher	Windows Service / Background Service	Continuously monitors designated network folder for new Excel files
Data Processing	Server-side (C#)	Reads Excel files, validates structure, extracts/matches IDs, deduplicates
Export Engine	EPPlus / ClosedXML	Generates Excel (.xlsx) and CSV exports of report data

Architecture Diagram (Logical Flow):

```

Network Folder (Excel files land here)
    |
    v
File Watcher Service (monitors folder 24/7)
    |
    v
Data Processor (read Excel, validate, extract IDs, deduplicate)
    |
    v
SQL Server Database (normalized tables)
    |
    v
Comparison Engine (match Evac vs AlcoConnect by extracted ID)
    |
    v
ASP.NET Web App (Date/Site selection -> 3-group report -> Export)

```

4. Automated File Monitoring

4.1 Folder Watcher Service

- Runs as a Windows Service or IIS-hosted background service
- Continuously monitors a configurable network folder path for new .xlsx files
- Scan interval must be configurable (e.g., every 1 minute, 5 minutes, etc.)

- No manual upload screen is necessary - automated monitoring is the only import path

4.2 File Detection & Classification

- Automatically detect whether a new file is an Evac Report or AlcoConnect report based on file name pattern or column structure
- Evac files: 1 per site per day
- AlcoConnect files: up to 12 per site per day
- Log each file detection: file name, type, timestamp, row count, processing status

4.3 Error Handling

- If a file has invalid structure (missing expected columns), log the error and skip the file
- If a file cannot be read (corrupted/locked), retry after next scan interval
- All errors logged for admin review

5. Data Processing & Storage

5.1 Data Import

- Read each Excel file and validate column structure matches expected schema
- Parse all data types correctly (dates, numbers, text)
- Extract the numeric portion from IMAOpenINX for matching (strip all letters, hyphens, prefixes/suffixes)
- Store the extracted numeric ID alongside the original value

5.2 Deduplication

When the AlcoConnect multi-import file contains rows already held in the database (same Staff ID + same Date + same Site), the system must keep the first version and silently ignore the duplicates. No overwriting.

5.3 Database Schema (Normalized)

The SQL Server database must include at minimum the following tables:

Table: EvacRecords

Column	Details
id	Primary key (auto-increment)
workgroup	From Excel
name	Employee name
organisation	Company name
roster_date	Date of roster entry
work_site	Site name
work_status	Shift/status
room	Room assignment
mobile	Phone number
ima_open_inx	Original IMAOpenINX value
extracted_id	Numeric ID extracted from IMAOpenINX
file_name	Source file name
imported_at	Timestamp of import

Table: AlcoConnectRecords

Column	Details
id	Primary key (auto-increment)
site	Site name
staff_id	Staff ID from file
staff_name	Employee name
job_title	Job title
phone	Phone number
email	Email address
manager	Manager name
machine_type	Breathalyser device model
test_date	Date of test
test_time	Time of test
result	BAC test result
serial_number	Device serial number
location	Test location on site
file_name	Source file name
imported_at	Timestamp of import

Table: FileImportLog

Column	Details
id	Primary key
file_name	Name of imported file
file_type	Evac or AlcoConnect
row_count	Number of rows imported
duplicates_skipped	Number of duplicate rows skipped
status	Success / Error
error_message	Error details if failed
imported_at	Timestamp

6. Comparison Engine

The comparison engine is the core logic of the system. It takes a selected Date and Site, queries both Evac and AlcoConnect tables, and produces three groups of results.

6.1 Matching Process

- Query all Evac records for the selected date and site
- Query all AlcoConnect records for the selected date and site
- Match records using the extracted numeric ID from Evac (extracted_id) against AlcoConnect Staff ID
- Classify each record into one of three groups

6.2 Three Report Groups

Group	Meaning	Business Implication
Evac Only	Employee is on the Evac roster but has NO matching breathalyser test	COMPLIANCE GAP - employee on site but not tested
AlcoConnect Only	Employee has a breathalyser test but is NOT on the Evac roster	Unmatched record - tested but not on roster
Matched	Employee appears in both Evac and AlcoConnect	Fully compliant - on roster AND tested

7. Web Interface & Reporting

7.1 User Workflow

The user interface is simple and focused:

- Step 1: Select a Report Date (date picker)
- Step 2: Select a Site (dropdown of available sites)
- Step 3: Click "Generate Report" button
- Step 4: View results displayed in 3 clearly separated groups
- Step 5: Export any group or all data to Excel/CSV

7.2 Report Group 1: Evac Only

Employees on the Evac roster but NOT found in AlcoConnect for that date/site.

Columns to Display:

Column	Source
staff_id (extracted)	From Evac extracted_id
Name	From Evac
Workgroup	From Evac
Organisation	From Evac
Work Site	From Evac
Work Status	From Evac
Room	From Evac
IMAOpenINX	From Evac (original value)
Mobile	From Evac

7.3 Report Group 2: AlcoConnect Only

Employees in AlcoConnect but NOT found in the Evac roster for that date/site.

Columns to Display:

Column	Source
Staff ID	From AlcoConnect
Staff Name	From AlcoConnect
Test Count	Count of tests for this employee on this date
First Test Time	Earliest test time
Last Test Time	Latest test time
Result	BAC result
Location	Test location on site

7.4 Report Group 3: Matched Records

Employees found in BOTH Evac and AlcoConnect for that date/site.

Columns to Display:

Combined data from both files showing key fields from each source for verification.

7.5 Export Functionality

- Export to Excel (.xlsx format) - each group on a separate sheet or selectable
- Export to CSV
- Export should include all columns and data visible in the report
- Export file name should include date and site (e.g., "Compliance_Report_Dalgaranga_2026-03-16.xlsx")

8. Configuration & Administration

8.1 Configurable Settings

Setting	Details
Watch Folder Path	Network folder path to monitor for new Excel files
Scan Interval	How often the watcher checks for new files (e.g., every 1-5 minutes)
Database Connection	SQL Server connection string
Site Names	List of valid site names (3 sites currently)

Settings should be manageable via a configuration file (appsettings.json) or a simple admin screen in the web application.

8.2 Logging & Audit Trail

- Log all file imports (file name, type, row count, duplicates skipped, timestamp)
- Log all errors (file read failures, validation errors, processing errors)
- Logs viewable via the web application or log files

9. Deliverables Checklist

- ☐ IIS-hosted ASP.NET web application with Date/Site selection interface
- ☐ 3-group reporting system (Evac Only, AlcoConnect Only, Matched Records)
- ☐ Automated file monitoring background service (Windows Service or hosted)
- ☐ SQL Server database with normalized schema
- ☐ Stored procedures / ETL scripts for data processing and comparison
- ☐ Excel file parser with column validation
- ☐ ID extraction logic (strip non-numeric from IMAOpenINX)
- ☐ Deduplication logic for multi-import AlcoConnect files
- ☐ Export functionality (Excel .xlsx and CSV)
- ☐ Configuration file for watch folder path and scan interval
- ☐ File import logging and error handling
- ☐ Complete source code
- ☐ Deployment guide (step-by-step for IIS + SQL Server setup)

[] 1 month post-deployment support (installation, training, bug fixes)

10. Project Milestones

Milestone 1: Database Design & Setup

- Finalize all requirements with client
- Design normalized SQL Server schema (tables, relationships, indexes)
- Create stored procedures for data import and comparison logic
- Set up development environment

Milestone 2: File Monitoring & Data Processing

- Build the file watcher service (monitors network folder)
- Implement Excel file reader with column validation
- Implement file type detection (Evac vs AlcoConnect)
- Build ID extraction logic (strip non-numeric from IMAOpenINX)
- Implement deduplication for multi-import files
- Build file import logging

Milestone 3: Comparison Engine & Reports

- Build the comparison engine (match Evac vs AlcoConnect by extracted ID)
- Generate 3 report groups (Evac Only, AlcoConnect Only, Matched)
- Build report data queries with date and site filtering

Milestone 4: Web UI - Part 1

- Build the ASP.NET web application shell
- Implement Date selector and Site selector interface
- Display 3-group report results on the web page
- Clean, easy-to-read table layout for each group

Milestone 5: Web UI - Part 2 & Export

- Implement Excel export (.xlsx) with proper formatting
- Implement CSV export
- UI polish and usability refinement
- Configuration settings screen or file

Milestone 6: Testing & Deployment

- End-to-end testing with sample files from all 3 sites
- Verify ID matching accuracy across file formats
- Test deduplication with multiple AlcoConnect file imports
- Test export functionality
- Deploy to client IIS server
- Write deployment guide
- Client handover and training

11. Future Enhancements (Phase 2)

The following features are out of scope for the MVP but should be considered for Phase 2 after the core system is proven and deployed:

- Advanced filtering: by Organisation, Workgroup, Work Status
- Historical trending and analytics (compliance rates over time)
- Custom report builder (user-defined columns and filters)
- Automated alerts and notifications (email reports on schedule)
- Scheduled email delivery of daily compliance report
- Additional data source integrations
- Performance dashboards with charts and summary statistics
- Light UI polish and branding

The MVP architecture should be built in a way that does not block these future enhancements.

12. Important Notes

- **Platform:** ASP.NET hosted on the client's existing IIS infrastructure with SQL Server database.
- **Sites:** 3 mining sites currently. Each site produces its own set of daily Evac and AlcoConnect files.
- **File Volume:** Approximately 150 rows per Evac file per day. AlcoConnect files can arrive up to 12 times per day per site.
- **Import Method:** Automated folder monitoring ONLY. No manual upload interface is required.
- **Deduplication:** When AlcoConnect multi-import files contain rows already in the database (same Staff ID + Date + Site), keep the first version and silently ignore duplicates.
- **ID Matching:** IMAOpenINX in Evac contains letters + numbers (e.g., "RAM-a25099A"). The system must strip all non-numeric characters to extract the ID and match against AlcoConnect Staff ID.
- **Hosting:** Client has existing IIS infrastructure. Application must be IIS-ready.
- **Source Code:** Complete source code must be delivered to the client.
- **Post-Launch Support:** 1 month of comprehensive post-deployment support including installation assistance, training, and bug fixes.
- **Scalability:** Architecture should support future Phase 2 enhancements without major refactoring.